

Computational Environments and Toolchains

Science

May, 2026

Computational Environments and Toolchains (A02381)

Short Title:	Computational Env & Toolchains	
Faculty	Science and Computing	
Department	Science	
Cluster	IT	
Author	RFRISBY	
Credits	5	Level: Intermediate

Description of Module / Aims

This is a practical module that requires the student to build, configure and manage computer systems and environments necessary for modern technologies.

Programmes

COMP-0580 BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)

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Indicative Content

- Linux overview: OS fundamentals; distributions; commonly used applications & tools, shell scripting
- Operating systems resources and virtualisation: file systems, virtual machines, hypervisors, resource and configuration management
- Introduction to computational science: survey of various computational tools (e.g. Python-based) used in exploring and analysing data in physics
- Networking services: IP addressing, DNS, DHCP, NAT
- Security: encryption, authentication, public key infrastructure
- Concepts and applications of parallel and distributed programs: process interaction using shared variables and message passing; parallel computing; development of correct programs; general problem solving techniques; scientific computing; distributed systems; programming on multiple processor machines

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Build, configure and manage secure network infrastructure services.
2. Use the Unix/Linux terminal interface, such as for data retrieval, text file manipulation, and scripting.
3. Use a variety of Web-based and other computational resources and applications for retrieving and analysing data from scientific repositories (Open Access Directory).
4. Develop solutions to problems in the retrieval and analysis of scientific data using modern programming languages (such as Python).
5. Construct a pipeline of different programs that automates collation, analysis, and reporting of data.

Learning and Teaching Methods

- This module adopts a practical approach with weekly hands-on laboratories and simulation exercises. Extensive use will be made of virtualised environments and microcontroller hardware in the practical work.
- This module will be delivered as two double lab-based sessions.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	40%	1,2,3
Assignment	60%	4,5

Assessment Criteria

<40%: Very little knowledge and understanding of Computational Environments and Toolchains. Unable to carry out critical thinking or practical skills.

40%-49%: Demonstrate a limited knowledge of Computational Environments and Toolchains. May have some problems with critical thinking and problem solving. Adequate practical skills.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within Computational Environments and Toolchains. Good practical skills.

60%-69%: Demonstrate skilled and sound knowledge of the subject matter. Show ability to analyse and logically discuss in an effective way. Good critical thinking and a high level of competence and efficiency in theoretical and practical areas of this subject.

70%-100%: All of the above to an excellent level. Demonstrate authoritative handling of material and ability to develop innovative solutions to problems that rise in the configuration and management of Computational environments.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Lab	36	
Independent Learning	87	

Supplementary Material

- "10 Simple Rules for the Care and Feeding of Scientific Data." <http://arxiv.org/pdf/1401.2134v1.pdf>
- "Best Practices for Scientific Computing." <http://arxiv.org/abs/1210.0530>
- Blum, R. and C. Bresnahan. *_Linux Command Line and Shell Scripting Bible_*. 3rd ed. New York: Wiley, 2015.
- Nemeth, E., G. Synder, T. Hein and B. Whaley. *_UNIX and Linux System Administration Handbook_*. 4th ed. New York: Prentice Hall, 2010.
- Newman, M. *_Computational Physics (with Python)_*. New York: CreateSpace Independent Publishing Platform, 2012.

Requested Resources

- COMPUTER LAB: BYOD Lab